

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

September 2, 2024

PART 1 - Problem 1

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC MATH TOOLBOX] [CONTROL TOOLBOX]

The water level $y(t)$ in a tank is modelled by the differential equation

$$\frac{dy(t)}{dt} + y(t) = z(t)$$

where $z(t)$ denotes the inflow. The inflow is a function of a valve position, wich in turn is controlled by the control signal $u(t)$. The relation between the control signal $u(t)$ and the flow $z(t)$ is given by the differential equation:

$$\frac{d^2z(t)}{dt^2} + \frac{dz(t)}{dt} + z(t) = u(t)$$

Assignment

Your code shall solve the following problems:

- What differential equation relates the water level $y(t)$ to the control signal $u(t)$? Exploiting the **Laplace transform**, evaluate the transfer function from the control signal $u(t)$ [the input] to the water level $y(t)$ [the output]. **Save** the resulting expression in the **symbolic variable TF_uy**.
- **HINT:** Detailed help on how to use the `laplace()` function is available [here](#).
- Create the **tf()** object **Gs**, corresponding to the symbolic expression **Tuy**.
- **HINT:** Detailed help on how to use the `tf()` function is available [here](#).
- Properly using the command **lsim()**, calculate the output evolution correspondig to the input $u(t) = [2 + \sin(2t)] \cdot \mathbf{1}(t)$, where $\mathbf{1}(t)$ is, as usual, the unit step function. For using in the command `lsim()`, generate an array of time instants $t_k \in [0, 20]$ s, with $t_k = k \cdot T$, $k = 0, 1, \dots$, $T = 0.01$ s. **Save** the resulting output evolution in the **numerical array Yt**.
- **HINT:** Detailed help on how to use the `lsim()` command is available [here](#).
- **Determine** the **state-space realisation in Observation Form** of the dynamical system **Gs**.
- **Save** the resulting matrices in the **numerical matrices matA, matB, matC**, and **matD**.
- **HINT:** for the topic "*state-space realisation*" refer to [Part 4](#) of the course material.

Solution

Init

```
clear
close all
clc
```

From the ODEs to the Transfer Function

```
syms t s y(t) z(t) u(t)
syms Ys Zs Us
syms y0 z0 dotz0

y0 = sym(0);
z0 = sym(0);
dotz0 = sym(0);

ODE_yz = diff(y(t), 1, t)+y(t) == z(t);
ODE_zu = diff(z(t), 2, t)+diff(z(t), 1, t)+z(t) == u(t);

L_yz = laplace(ODE_yz);
L_yz = subs(L_yz, [laplace(y(t),t,s), laplace(z(t),t,s), y(0)], [Ys, Zs, y0])
```

$L_{yz} = Ys + Ys s = Zs$

```
TF_yz = solve(L_yz,Ys)/Zs
```

TF_yz =
 $\frac{1}{s + 1}$

```
L_zu = laplace(ODE_zu);
L_zu = subs(L_zu, [laplace(u(t),t,s), laplace(z(t),t,s)], [Us, Zs])
```

L_zu =
 $Zs - z(0) + Zs s - s z(0) - \left(\left(\frac{\partial}{\partial t} z(t)\right)\Big|_{t=0}\right) + Zs s^2 = Us$

```
L_zu = subs(L_zu, [z(0), subs(diff(z(t), t), t, 0)], [z0, dotz0])
```

$L_{zu} = Zs s^2 + Zs s + Zs = Us$

```
TF_zu = solve(L_zu,Zs)/Us
```

TF_zu =
 $\frac{1}{s^2 + s + 1}$

```
TF_uy = TF_zu*TF_yz
```

TF_uy =

$$\frac{1}{(s+1)(s^2+s+1)}$$

The Transfer Function and the Output Movement

```
Gs = tf(1,[1 1])*tf(1,[1 1 1])

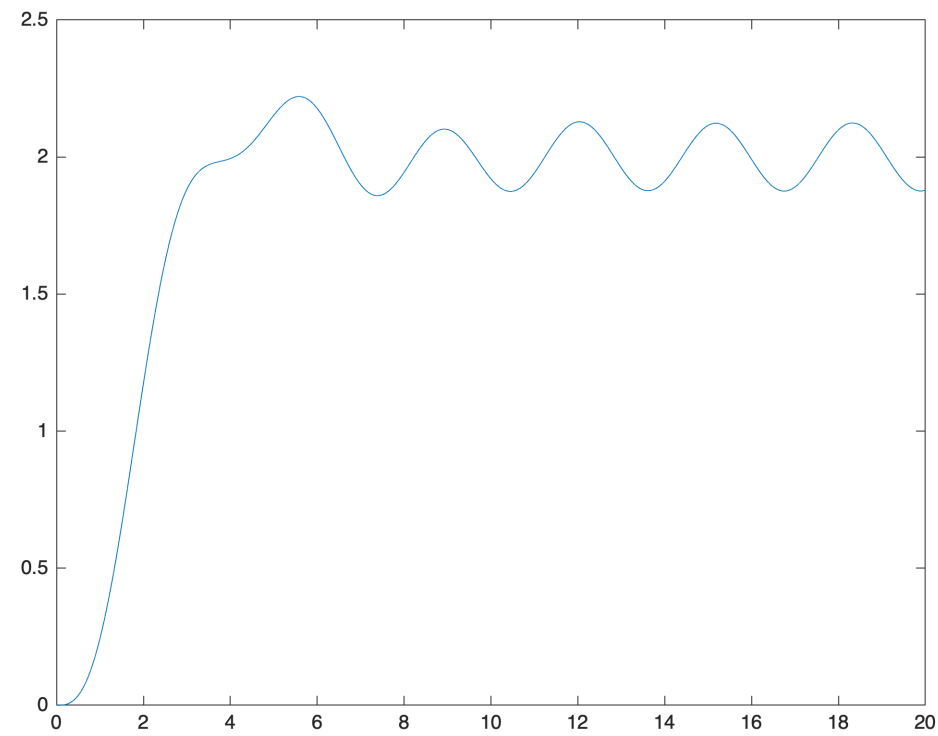
Gs =

      1
-----
s^3 + 2 s^2 + 2 s + 1

Continuous-time transfer function.
Model Properties

Tinit = 0; Tend = 20;
DeltaT = 0.01;
Tvect = (Tinit:DeltaT:Tend);
Uvect = 2+sin(2*Tvect);
Yt = lsim(Gs, Uvect, Tvect);

figure;plot(Tvect, Yt) % NON richiesto!!
```



The State-Space Realisation

```
% the arrays with the numerator & denominator coefficients
```

```
[NsTFcoeffs, DsTFcoeffs] = tfdata(Gs, 'v')
```

```
NsTFcoeffs = 1x4
    0    0    0    1
DsTFcoeffs = 1x4
    1    2    2    1
```

```
n = numel(pole(Gs)); % how many state variables? <--> the degree of denominator of G(s)
m = numel(zero(Gs)); % the degree of N(s)
NsTFcoeffs = NsTFcoeffs(end-m:end);
```

```
% check
assert(n==(numel(DsTFcoeffs)-1))
assert(m==(numel(NsTFcoeffs)-1))
```

```
B2mat = sym([NsTFcoeffs; zeros(n-m-1, 1)]); % the B matrix
```

```
C2mat = sym([zeros(1, n-1), 1]); % the C matrix
```

```
lastCol_A2 = transpose(DsTFcoeffs(1:end-1));
A2mat = sym([zeros(1, n-1); eye(n-1)], ...
            -lastCol_A2]); % the A matrix
```

```
matA = double(A2mat);
matB = double(B2mat);
matC = double(C2mat);
matD = 0; % <-- SISO stricly proper system!
```

```
% last check
LTIsys = ss(matA, matB, matC, matD)
```

```
LTIsys =
```

```
A =
      x1  x2  x3
x1      0   0  -1
x2      1   0  -2
x3      0   1  -2
```

```
B =
      u1
x1      1
x2      0
x3      0
```

```
C =
      x1  x2  x3
y1      0   0   1
```

```
D =
      u1
y1      0
```

```
Continuous-time state-space model.
Model Properties
```

```
% NON richiesto
LTIsys_TF = tf(LTIsys)
```

LTIsys_TF =

$$\frac{1}{s^3 + 2 s^2 + 2 s + 1}$$

Continuous-time transfer function.
Model Properties

```
% NON richiesto
% check equivalence -- is L_inf norm of difference (error) close enough to zero?
tol = 1e-10;
assert( norm( LTIsys_TF - Gs , 'inf' ) < tol , "G is not correct.")
```